

Apratim Phadke

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About

E&TC undergraduate at VIIT Pune with hands-on experience in RTL design, Physical Design (PnR), STA, and custom-cell layout. Completed RTL-to-GDS implementation on Sky130 PDK, achieving timing closure and clean DRC/LVS verification. Designed a 6T SRAM cell in Cadence Virtuoso. Built a quad-core RISC-V processor in SystemVerilog with hybrid TAGE-Perceptron prediction, improving branch prediction accuracy over conventional schemes. Experienced in custom layout, timing optimization, and PPA-driven Physical Design across digital VLSI projects.

Projects

8-bit SAR ADC — Full RTL-to-GDS (PnR + STA) | *Verilog, Yosys, OpenROAD, Magic, KLayout, Sky130*

- Executed complete PD flow on Sky130: synthesis (Yosys), floorplan (core utilization 58%, $120 \times 120 \mu\text{m}$), PDN, placement (847 std cells), CTS (32 clock buffers inserted), and routing.
- Performed STA (setup/hold analysis, clock uncertainty, false-path and multicycle-path exceptions via SDC); achieved timing closure at 50 MHz — WNS +0.08 ns, TNS 0 ns post-CTS.
- OpenROAD power estimation: dynamic 1.2 mW, leakage 48 μW ; routing congestion <5% overflow; DRC: 0 violations, LVS: clean post sign-off in Magic/KLayout.
- Built pyuvvm + Cocotb UVM environment with SVA assertions and functional coverage; validated SAR logic, DAC feedback, and comparator under constrained-random stimulus.

4-bit 6T SRAM — Full-Stack Design (Schematic to FPGA) | *Cadence Virtuoso, Microwind, Verilog, Xilinx Vivado*

- Designed a 6-transistor SRAM bitcell in Cadence Virtuoso (90 nm): schematic entry, DC operating point and transient simulation (read/write margins verified), followed by full custom layout with DRC/LVS sign-off.
- Prototyped the same cell in Microwind for CMOS process visualization and cross-validated layout parasitics; characterized read SNM and write margin across process corners.
- Implemented 4-bit SRAM array (4 rows $\times$ 1-bit cell, address decoder, sense amplifier, write driver) in Verilog; synthesized and implemented on Nexys 4 DDR (Artix-7 FPGA) via Xilinx Vivado with full timing closure.

RONAN: Neural Network-Enhanced RISC-V Processor | *SystemVerilog, Cocotb, Verilator, Python*

- Quad-core RISC-V with hybrid TAGE-Perceptron branch prediction: 99.89% accuracy, 0.999 IPC vs. 0.508 baseline across 5M+ cycles; synthesized on Nexys 4 DDR (Artix-7).
- Parameterized Cocotb/Verilator testbenches with SVA assertions and constrained-random stimulus for perceptron table, GHR, and jog-packet unit across 5 structured workloads.

AMBA SoC with UART & I2C Peripherals | *Verilog, Xilinx Vivado, Artix-7 FPGA (Nexys-4 DDR)*

- Designed and verified a complete SoC using AMBA APB v3 with UART and I2C controllers; achieved 5.12% LUT utilization through optimized RTL and comprehensive testbench validation.

Skills

Physical Design & STA: OpenROAD (Floorplan, Placement, CTS, Routing), Yosys, SDC Constraints Development, Setup/Hold Analysis, Clock Uncertainty, False Paths, Multicycle Paths, Timing Closure, DRC/LVS, PPA Optimization

Power Analysis: OpenROAD Power Estimation, Dynamic & Leakage Power, PPA-aware Design, PrimeTime PX (familiar)

CDC & Clock: CDC Analysis, Synchronizer Design, Metastability Mitigation, Multi-clock Domain Design

Industry EDA (familiar): Synopsys VCS, Design Compiler, IC Compiler II, PrimeTime, StarRC, Formality, Verdi, VC SpyGlass

Other EDA Tools: Cadence Virtuoso, Microwind, ModelSim, Icarus Verilog, Xilinx Vivado, Quartus Prime, Magic, KLayout

HDL: Verilog, SystemVerilog

Verification: UVM (pyuvvm), Cocotb, SVA (Assertions), Functional Coverage, Constrained Random Verification, GTKWave/Verdi

Programming: C/C++, Python, TCL, Shell Scripting

Protocols: UART, SPI, I²C, APB, CAN

Tools & OS: Linux, Git, GDB, STM32CubeIDE, MATLAB/Simulink

Education

Vishwakarma Institute of Information Technology, Pune

Bachelor of Technology, Electronics & Telecommunication Engineering

2023 – 2027

CGPA: 8.41

Publications

RONAN: A Neural Network-Inspired Pipelined RISC-V Processor with Perceptron-Based Branch Prediction

- A. Phadke, R. Hadkar, H. Satpute, S. Gondepatil — Supervised by Dr. Shailesh Kulkarni, VIIT Pune. *Under Review*, 2026.

Certifications

Fundamentals of Digital Design for VLSI Chip Design — L&T EduTech (2025) | **Introduction to Chip Design with**

Open-Source EDA Tools — Coursera (2025) | **VLSI Chip Design and Simulation with Electric VLSI EDA Tool** — L&T

EduTech (2025) | **Introduction to Microprocessors** — ARM (2025) | **Using Git for Distributed Development** — The Linux

Foundation (2025) | **Modeling and Simulation with Simulink** — MathWorks (2025)

Leadership Experience & Achievements

General Manager, IEEE VIIT Student Branch

October 2024 – Present

- Organized technical workshops on Python, IoT, and sensor integration for 100+ high school students; coordinated IEEE ICBDS Conference 2024.

IEEE IoT Hackathon — 2nd Prize | Smart India Hackathon 2025 (Internal) | DYPDPU Hackathon 1.0 (Feb 2025)

- Guided junior team to 2nd place at IEEE IoT Hackathon building a real-time embedded sensor monitoring system; participated in SIH 2025 and DYPDPU Hackathon.